# Alexis Lavail

PhD student in Astronomy

alexis.lavail@physics.uu.se

http://aalex.is

Address: Flogstavägen 77E-533
75272 Uppsala (Sweden)
Phone: +46(0)72 257 0369
Nationality: French
Personnummer: 19910526-3736


## Research experience

**PhD student in Astronomy** — **2013-present**
Uppsala University, European Southern Observatory — Uppsala (Sweden), Garching (Germany)

Thesis title: *Magnetic fields of cool stars from near-infrared spectropolarimetry*
Supervisors: **Oleg Kochukhov**, **Nikolai Piskunov**, **Gaitee Hussain**
Planned defense date: **April/May 2020**
Licentiate degree passed: May 2019

> **My research focuses on studying the magnetic fields of pre main-sequence stars and M-dwarfs** from high-resolution near-infrared spectroscopy and optical spectropolarimetry. Additionally, I participate in **instrumentation work on the polarimeter unit for CRIRES+**: the upgrade of the CRIRES high-resolution near-infrared spectrograph for the Very Large Telescope.

**European Space Agency Stagiaire** — 2013, 4 months
European Space Agency ESTEC — Noordwijk (Netherlands)

**Master degree research project** — 2012, 2 months
Uppsala University — Uppsala (Sweden)

## Education

**Master degree in astronomical and space-based engineering** — 2013
Observatoire de Paris — Paris, Meudon, Orsay (France)

**Bachelor degree in physics** — 2011
Université Paul Cézanne Aix-Marseille III — Marseille (France)

## Refereed publications

- **Lavail, A.**, Kochukhov, O., Hussain, G.: *Characterising the surface magnetic fields of T Tauri stars with high-resolution near-infrared spectroscopy*, 2019, accepted for publication in A&A
- **Lavail, A.**, Kochukhov, O., Wade, G. A.: *A sudden change of the global magnetic field of the active M dwarf AD Leo revealed by full Stokes spectropolarimetric observations*, 2018, MNRAS, 479, 4836L
- **Lavail, A.**, Kochukhov, O., et. al: *Magnetic fields of intermediate mass T Tauri Stars*, 2017, A&A, 608, A77
- Kochukhov, O., **Lavail, A.**: *The global and small-scale magnetic fields of fully convective, rapidly spinning M dwarf pair GJ65 A and B*, 2017, ApJ, 835, L4

## Computer skills

- Good overall knowledge of **GNU/Linux** operating system and environment. Familiar with **macOS**.
- Experienced in **IDL/GDL** and **Python** programming.
- Fluent in **HTML/CSS** and **Markdown** languages for wiki, documentation, or webpages editing.

## Languages

French: native — English: fluent — Swedish: intermediate — Spanish: intermediate

## Conferences, Presentations, and Workshops

- Astronomdagarna 2019 Poster presentation — Stockholm University (Sweden) Oct 2019
- STARRY final conference Contributed talk — Leeds (United Kingdom) Jun 2019
- Astronomy department seminar Seminar presentation — Université de Montpellier (France) Apr 2019
- Cool Stars 20 Poster presentation — Boston, MA (USA) Jul 2018
- Astronomdagarna 2017 Contributed talk and poster presentation — IRF Kiruna (Sweden) Sep 2017
- Cool Stars 19 Poster presentation — Uppsala (Sweden) Jun 2016
- BCool collaboration meeting Contributed talk — Vienna Observatory (Austria) Feb 2016
- Astronomdagarna 2015 Poster presentation — Uppsala University (Sweden) Oct 2015

## Accepted observing proposals

- **Lavail, A.**, Kochukhov, O., and Wade, G. Instrument: ESPaDOnS at CFHT (Hawaii), **4 hours**, service mode. *Tracking the unprecedented magnetic evolution of the magnetic field of the active M dwarf AD Leo.* 2019
- **Lavail, A.**, Kochukhov, O., Hussain, G., and Piskunov, N. Instrument: GIANO at TNG (La Palma), **26 hours**, visitor mode. *Near-infrared magnetic diagnostics for cool low-mass active stars.* 2017
- **Lavail, A.**, Wade, G., Kochukhov, O. Instrument: ESPaDOnS at CFHT (Hawaii), **23 hours**, service mode. *A search for Zeeman linear polarization in spectral lines of active M dwarfs.* 2016

## Service & duties

### Conference organization

- Uppsala/Stockholm astronomy PhD workshop http://aalex.is/uusu3 — Norreda Torp (Sweden) Jan 2019
- Cool Stars 19 http://coolstars19.com — Uppsala (Sweden) Jun 2016
- Astronomdagarna 2015 http://www.astro.uu.se/astro2015 — Uppsala University (Sweden) Oct 2015

### Duties & outreach

- Production of printed/online posters for outreach astronomy events 2016-present
- Responsible for the astronomy division webpages at Uppsala University 2014-2018
- PhD representative at the astronomy division council at Uppsala University 2015-2017

## Grants and stipends

- General announcement for Astronomy and space science 2018
  Amount: 30 000 SEK — KVA (Royal Academy of Sciences)
  Project: *Organization of a joint workshop for PhD student in astronomy and space physics at Stockholm and Uppsala Universities.*
- Liljewalch resestip (Liljewalch travel stipend) 2018
  Amount: 38 000 SEK — C.F. Liljewalchs stipendiestiftelse
  Project: *Research visit at Université de Montpellier, attendance to Cool Stars 20 and SunStars school*
- ESO DGDF (PI: Gaitee Hussain) 2017
  Amount: 3 000 € — ESO Director General Discretionary Funding
  Project: *Research visit at the European Southern Observatory HQ in Garching.*

## Interests

datavizualisation, inclusiveness in science, hiking, bouldering, cross-country skiing, beer-brewing, baking, world literature and music, science-fiction, dystopia, web and poster design, typography ...